

Cafe Analytics: Revenue Intelligence Report

Item Performance, Payment Behavior & Temporal Sales Analysis

Sector: Food & Beverage Retail
Team: Group 17 | Section A
Newton School of Technology

Data Visualization & Analytics — Capstone Project | 2024-2028

Context: The Thin Margin Reality

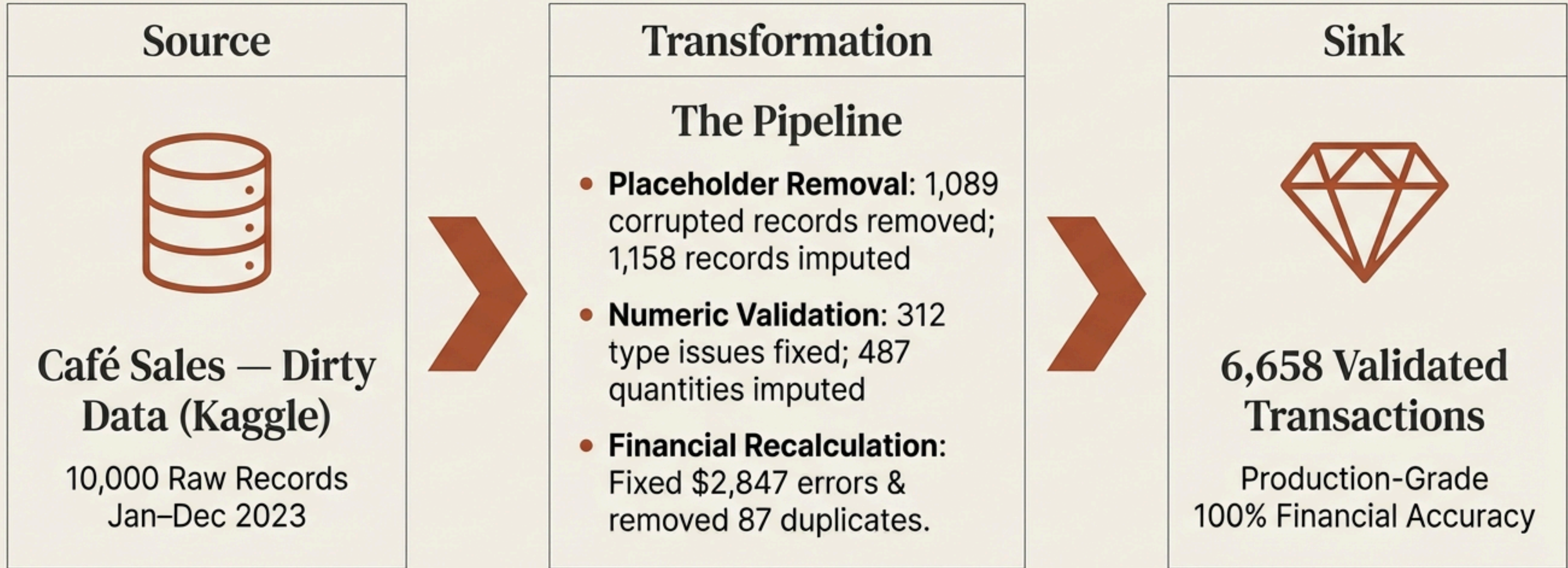
The F&B retail sector operates on high velocity and thin margins. Profitability hinges on three operational levers: Menu Engineering, Operational Throughput, and Payment Infrastructure.

The Problem: A Revenue Blind Spot

The café captures rich POS data but lacks governance. 33.42% of raw records were corrupt or invalid. Decisions were based on inflated revenue (\$2,847 in calculation errors) and duplicate transactions.

Research Question: How do item performance, payment preferences, and time-based sales trends impact overall café revenue, and which factors drive peak sales periods?

The Data Refinery: Pipeline Flow



Data Dictionary Key Fields: transaction_id (PK), total_spent (Verified), transaction_month (Engineered)

KPI & Metric Framework

Structuring success across four dimensions.

Revenue Dimension • Total Revenue (Primary Benchmark) • Average Transaction Value (Basket Size)	Product Dimension • Revenue per Item (Profitability) • Item Sales Volume (Pareto Analysis)
Payment Dimension • Payment Method Share (Cash vs. Digital) • Avg. Spend by Method (Behavior Analysis)	Operational Dimension • In-Store vs. Takeaway Split • Peak Revenue Months (Staffing)

Exploratory Insights: The Fundamental Truths

The Revenue Engine 26.4%

Heldane Display of Total Revenue is Salad

Salad generates \$15,963.50 — 26.4% of total annual revenue. Its simultaneous leadership in both unit price (\$4.27) and volume (3,732 units) creates a revenue concentration risk. Any supply disruption, ingredient cost spike, or competitor substitution poses a disproportionate revenue threat.

The Volume Trap 33%

Volume from Low-Value Items

Cookie, Tea, and Coffee collectively account for 33.1% of total units sold (6,743 units) but only 17.8% of revenue (\$10,777). These items attract foot traffic but underperform on revenue contribution — a classic menu engineering challenge.

Channel Reality 70.3%

Revenue from In-Store

With \$42,522.00 generated in-store versus \$17,961.50 takeaway, the café's primary value driver is clearly the dine-in experience. However, the per-transaction gap is small (\$9.15 vs. \$8.94), confirming that takeaway underperformance is a volume problem — not a spending-level problem.

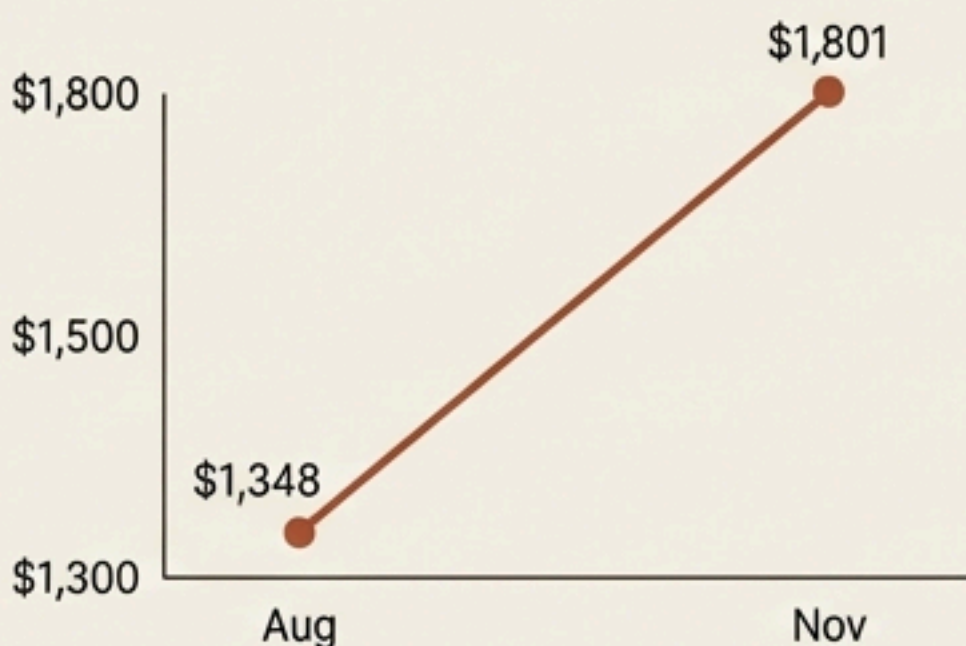
Payment Myth \$9.00

Identical Spend Across Methods

Cash (\$9.15 ATV), Digital Wallet (\$9.07), and Credit Card (\$9.02) transactions are statistically equivalent in average spend. This means the payment method does not influence how much a customer spends — selection should instead be guided by transaction cost and processing efficiency.

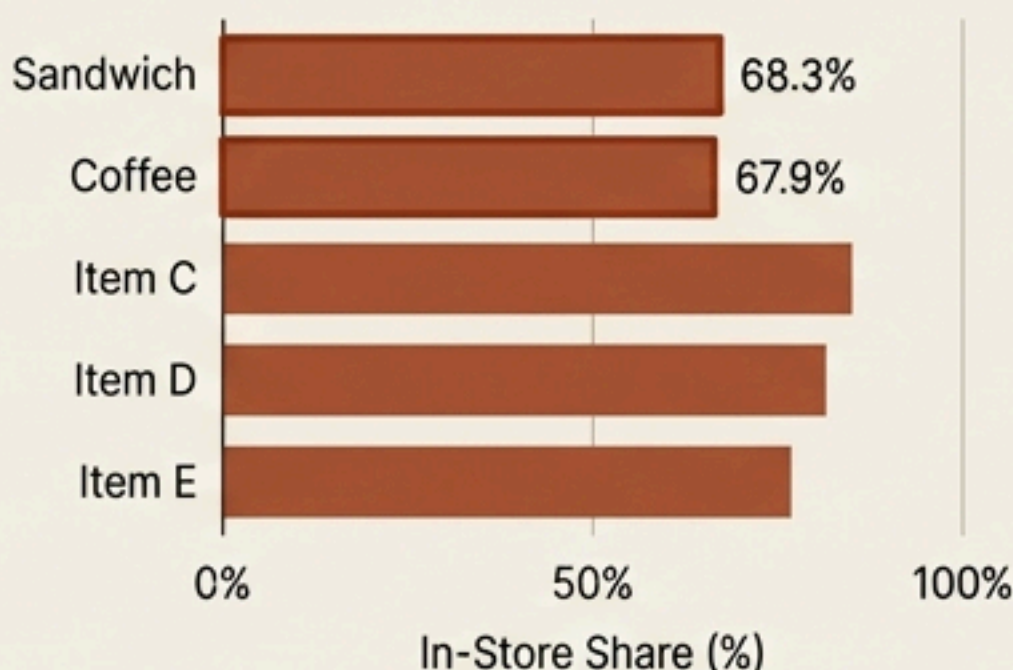
Advanced Analysis: Trends & Segmentation.

The Digital Shift (Q4)



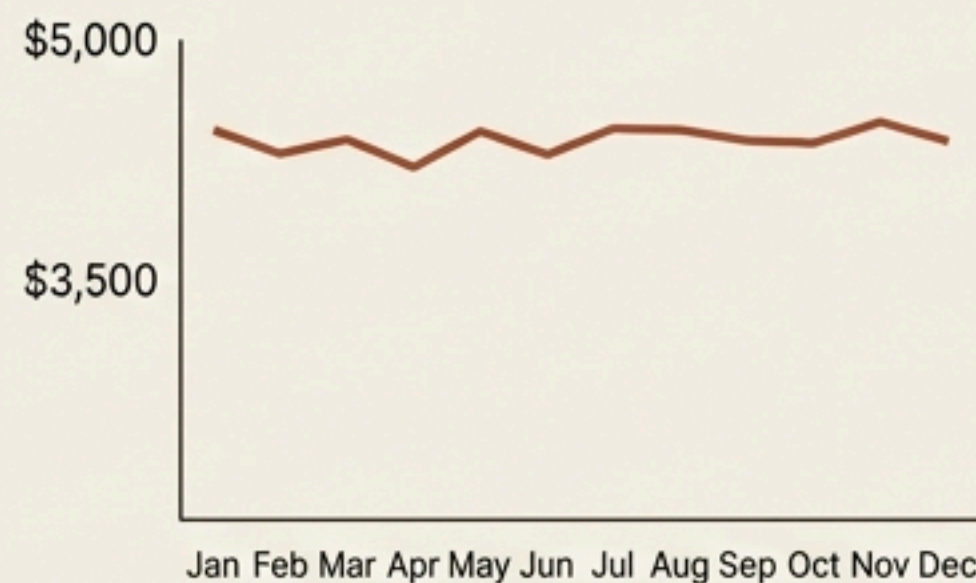
Digital Wallet revenue grew 33.6% from August to November (\$1,348.50 → \$1,801.50). By November, Digital Wallet is within \$110 of Cash for that month, signalling a structural shift in consumer payment preference accelerating toward year-end.

Takeaway Affinity



Sandwich and Coffee have the lowest in-store revenue share (68.3% and 67.9% respectively), meaning a greater proportion of their sales occurs through takeaway than for any other items. These are natural candidates for takeaway-specific bundle promotions and packaging investment.

Revenue Stability



The peak-to-trough revenue range of \$738.00 (16.2% variance) across 12 months indicates structurally stable demand. The café does not experience dramatic seasonal swings, which simplifies inventory planning, staffing, and financial forecasting.

Dashboard Walkthrough: Executive View

\$60,483.50

Total Verified Revenue (2023)

\$9.09

Average Transaction Value

6,658

Validated Transactions

Filters

Month

All Months



Location

All Locations



Payment Method

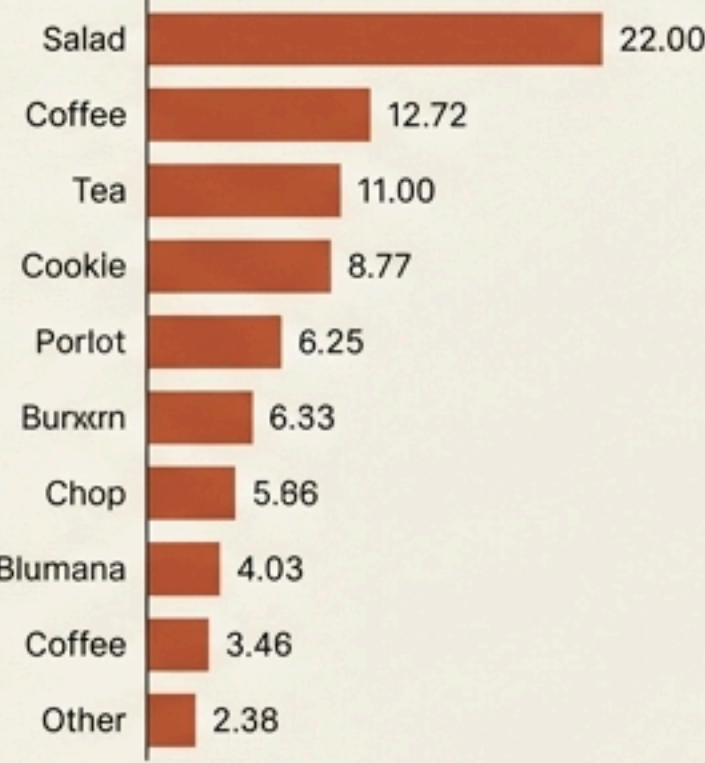
All Methods



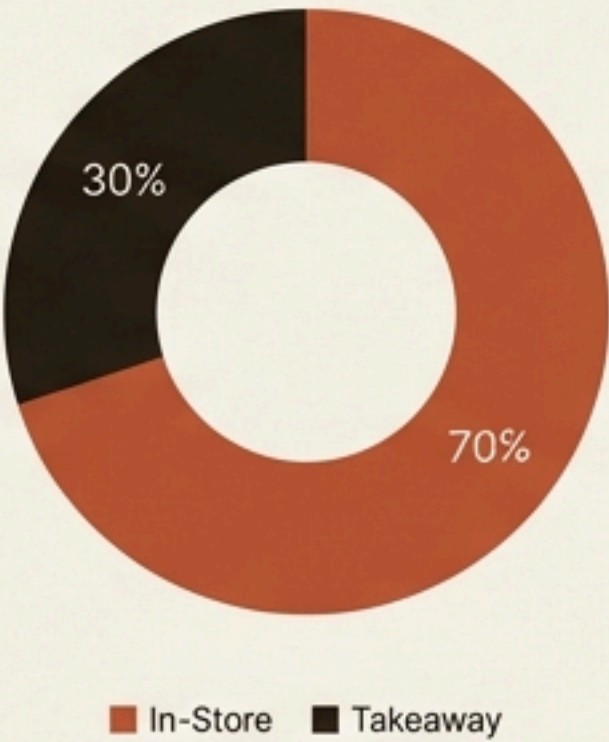
Monthly Revenue Trend



Item Performance



Channel Split



Strategic Recommendations.

High Priority

Diversify Salad Risk

Introduce a premium Salad variant (e.g., seasonal or protein-enhanced) at a 20–25% price premium while simultaneously promoting Smoothie and Sandwich as secondary revenue anchors through combo pricing. Reducing revenue concentration below 22% for any single item improves financial resilience.

Medium-High Priority

Monetize Footfall

Introduce “Add-On” pairings for Cookie, Tea, and Coffee (e.g., Cookie + Coffee at a 10% bundle discount). If 20% of the 6,743 units sold across these items convert to a bundle adding \$2.00 in revenue, this generates approximately \$2,697 in incremental annual revenue.

Medium-High Priority

Digital Takeaway

Digital Wallet is already the leading payment method in the takeaway channel. Ensuring zero-friction digital payment acceptance at the takeaway counter — including NFC terminals and QR code payment options — directly supports the channel growth trajectory and reduces cash-handling costs.

High Priority

Staff for Peaks

These three months (January, March, and June) generate 26.5% of annual revenue despite representing 25% of the year. Ensure full inventory availability, reduced stockout risk, and optimised staffing ratios during these periods. A 5% operational efficiency gain in peak months recovers approximately \$792 in revenue.

Impact & Value Creation.

~\$14,036

Total Annual Value Creation

Revenue Uplift (+\$12,236)

- Salad Premium Upsell: +\$6,717
- Bundle Conversion: +\$2,697
- Peak Efficiency Gains: +\$792
- Other Initiatives: +\$2,030

Cost Savings (~\$1,800)

- Reduced cash handling costs
- Overtime optimization

Value Delivered: From raw, unstructured POS data to seven actionable business recommendations with a quantified \$14,036 annual value creation opportunity — achieved through rigorous data engineering, systematic analytics, and business-language interpretation of every finding.

Limitations & The Horizon

Current Constraints

- Single Year Scope (2023 only, seasonality unconfirmed)
- No Timestamps (Monthly granularity only)
- Revenue Only (No COGS/Profit margin data)

Future Scope (Next Steps)

1. **Timestamp Integration**: Collect hourly data for labor heatmaps.
2. **Profit Intelligence**: Integrate COGS to measure Gross Margin.
3. **Customer Cohorts**: Track IDs for Retention & LTV analysis.
4. **Forecasting**: Build ARIMA models for inventory planning.